



```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from glob import glob
import IPython.display as ipd
from tqdm.notebook import tqdm
```

```
In [ ]: #import subprocess
```

```
In [ ]: ipd.Video("026c7465-309f6d33.mp4",width = 200)
```

Out[]: Your browser does not
support the video
element.

```
In [ ]:
```

Open the Video and Read Metadata

```
In [ ]: !pip install opencv-python
```

Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)

```
In [ ]: # Load in video capture
import cv2
cap = cv2.VideoCapture('026c7465-309f6d33.mp4')
```

```
In [ ]: # Total number of frames in video
cap.get(cv2.CAP_PROP_FRAME_COUNT)
```

Out[]: 2398.0

```
In [ ]: # Video height and width
height = cap.get(cv2.CAP_PROP_FRAME_HEIGHT)
width = cap.get(cv2.CAP_PROP_FRAME_WIDTH)
print(f'Height {height}, Width {width}')
```

Height 720.0, Width 1280.0

```
In [ ]: # Get frames per second
fps = cap.get(cv2.CAP_PROP_FPS)
print(f'FPS : {fps:0.2f}')
```

FPS : 59.94

```
In [ ]: cap.release()
```

Pull the image from the first frame

```
In [ ]: cap = cv2.VideoCapture('026c7465-309f6d33.mp4')
ret, img = cap.read() #will return TRUE everytime w
print(f'Returned {ret} and img of shape {img.shape}')
```

Returned True and img of shape (720, 1280, 3)

Loading the First Image

```
In [ ]: plt.imshow(img) #opencv loads color in BGR format, matplotlib lib expects RGB fo
```

Out[]: <matplotlib.image.AxesImage at 0x7faad2e85be0>



```
In [ ]: ## Helper function for plotting opencv images in notebook
def display_cv2_img(img, figsize=(10, 10)):
    img_ = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    fig, ax = plt.subplots(figsize=figsize)
    ax.imshow(img_)
    ax.axis("off")
```

```
In [ ]: display_cv2_img(img)
```



```
In [ ]: cap.release()
```

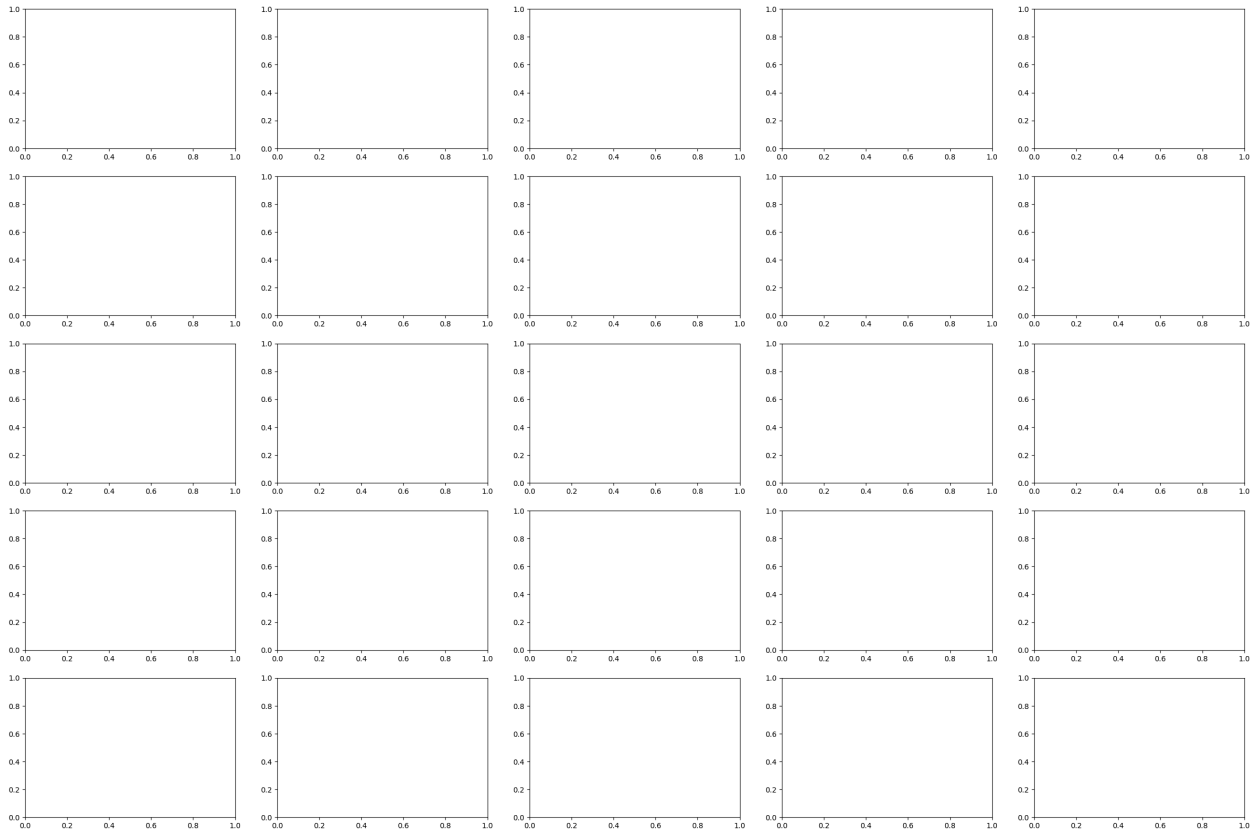
Display Multiple Frames from the Video

```
In [ ]: fig, axs = plt.subplots(5, 5, figsize=(30, 20))  
plt.show()
```



```
In [ ]: fig, axs = plt.subplots(5, 5, figsize=(30, 20))
        axs = axs.flatten()
        plt.show()
```

```
# Without flatten → 2D grid (row, column)
# With flatten → 1D list (single index)
```



```
In [ ]: import cv2
        fig, axs = plt.subplots(5, 5, figsize=(30, 20))
        axs = axs.flatten()

        cap = cv2.VideoCapture("026c7465-309f6d33.mp4")
        n_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

        img_idx = 0
        for frame in range(n_frames):
            ret, img = cap.read()
            if ret == False:
                break
            if frame % 100 == 0:                                     #frame is di
                axs[img_idx].imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
                axs[img_idx].set_title(f'Frame: {frame}')
                axs[img_idx].axis('off')
                img_idx += 1

        plt.tight_layout()                                         #automatical
        plt.show()
        cap.release()
```




```
In [ ]: labels = pd.read_csv('mot_labels.csv', low_memory=False)
        ##labels.head()
        #video_labels = (labels.query('videoName == "026c7465-309f6d33"').reset_index(
        #video_labels["video_frame"] = (video_labels["frameIndex"] * 11.9).round().ast
```

```
In [ ]: #print(video_labels.shape)
```

```
(0, 14)
```

```
In [ ]: print(labels["category"].value_counts())
```

```
category
car          394391
pedestrian   67862
truck        22812
bus          10694
bicycle       3823
other vehicle 3756
rider         3307
motorcycle    2277
other person   782
trailer        376
train          156
Name: count, dtype: int64
```

```
In [ ]: cap = cv2.VideoCapture("026c7465-309f6d33.mp4")
        n_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

        img_idx = 0
```

```
for frame in range(n_frames):  
    ret, img = cap.read()  
    if ret == False:  
        break  
    if frame == 1035:  
        break  
cap.release()
```

```
In [ ]: display_cv2_img(img)
```



```
In [ ]:
```